CardioTwin AI

A Multimodal Cardiovascular Digital Twin Platform for Proactive Care

Happiest Health Reimagining and Reforming Healthcare in India Summit — 2026

Digital Twin Challenge 2026, Bengaluru

Submitted by Team Phoenix:

• **Vansh Singla** (Team Leader, Full-Stack & Biomedical AI Lead)

• **Dev Garg** (ML Engineer & Signal Processing)

• **Prince Pal** (Healthcare Data Architect & Frontend Engineer)

College & Incubator: Maharaja Agrasen Institute of Technology (MAIT), Delhi

MAIT Innovation & Incubation Centre

1. Problem Statement: The Cardiovascular Crisis in India

Why proactive digital twin medicine is urgent in India

• **Mortality Leader:** Cardiovascular diseases (CVDs) account for **over 28% of all deaths in India** (ICMR, GBD Study).

• **The 'Silent Killer':** Hypertension affects over **200 million Indian adults**, with >60% unaware until an acute event strikes.

• **Reactive Healthcare Model:** Traditional cardiology is reactive—diagnosing patients only after symptoms, infarcts, or strokes occur.

• **The Digital Twin Solution:** Create a living, dynamic virtual patient replica that continuously simulates physiological state, fuses historical records with real-time wearable streams, and predicts adverse events days before they happen.

2. Dual-Stream Multimodal Data Ingestion

Fusing static clinical history with continuous wearable signals

STREAM 1: STATIC EHR (Past Medical Context)	STREAM 2: DYNAMIC WEARABLES (Real-Time Signals)
• Synthea Electronic Health Records engine.	• PhysioNet BIDMC PPG: 125 Hz optical plethysmogram.
• Demographics: Age, Gender, BMI.	• PhysioNet MIT-BIH ECG: 360 Hz Lead MLII recordings.
• Historical Diagnoses: Essential Hypertension, Type 2 Diabetes.	• Real-time Bluetooth Low Energy (BLE) stream state machine.
• Lifestyle factors: Tobacco use, Physical activity, Sleep duration.	• SciPy Butterworth bandpass filtering (0.5–5.0 Hz).
• Prior lab encounters & medications.	• Heart Rate Variability (HRV) & Signal Quality Index (SQI).

3. Predictive Models & Clinical Outcomes

Measurable artificial intelligence models deployed in CardioTwin AI

• **Multimodal Late-Fusion Engine:** Synthesizes physiological waveforms (40%), tabular vitals (30%), clinical NLP notes (20%), and report OCR scans (10%) into an interpretable 0–100 Cardiovascular Risk Index.

• **PhysioNet MIT-BIH Arrhythmia Classifier:** Supervised Random Forest classifier trained on gold-standard beat annotations (N, V, A, L, R). Achieves **71.74% overall accuracy** and **0.599 weighted F1** on strictly held-out subject data (zero participant leakage).

• **Blood Pressure Trend Forecasting:** Detects pre-hypertensive transitions and stage 2 arterial spikes from combined pulse transit metrics and structured vitals.

• **Explainable AI (XAI) Attribution:** Calculates mathematical modality contributions and flags specific clinical warnings rather than functioning as an opaque black box.

• **Longitudinal SQLite Health Memory:** Tracks patient trajectory over time, computing comparative Risk Deltas against prior visits.

4. Clinician-Ready Digital Twin Interface

A professional 3-column virtual patient experience

• **3D Translucent Digital Twin Avatar:** Vector SVG cardiac centerpiece featuring animated heart pulse, arterial tree, ribcage vectors, and DNA double-helix motifs.

• **4 Interactive Physiological Hotspots:**

1. *Trigeminal Autonomic:* Sinus rhythm & SA node firing.
2. *Posterior Hypothalamus:* Autonomic tone & Heart Rate Variability.
3. *Trigeminovascular System:* Arterial vascular compliance & BP regulation.
4. *Autonomic Reflex:* Capillary microcirculation & SpO2 oxygen saturation.

• **Doctor Interaction Modules:** Real-time vitals wave monitors, complaints documentation with biomedical NLP, checkup schedule timeline, and ReportLab 3.0 stamped medical PDF exporter.

• **Interactive Guided Tour:** Built-in step-by-step onboarding walkthrough with directional pointer arrows, spotlights, and radar beacons for first-time evaluators.

5. Data Ethics, Privacy & Regulatory Compliance

Navigating healthcare privacy laws (DPDP Act 2023 & HIPAA)

• **Strict Data Provenance Tagging:** Every record in the system is visibly and immutably tagged: OPEN_RESEARCH_DATA, SYNTHETIC_EHR, or AUTHORIZED_PATIENT_DATA.

• **Zero Silent Mock Data:** No fabricated patient profiles or simulated numbers are ever silently presented as genuine human data.

• **Local Data Persistence:** SQLite database runs locally on the clinical machine. No unauthorized data leaves the hospital or user boundary.

• **Responsible AI Notice:** CardioTwin AI is developed as an educational and research proof-of-concept for the Happiest Health Summit. It is designed to empower clinicians, not replace them.